SMART INDIA HACKATHON 2025

TITLE PAGE

- **PROBLEM STATEMENT ID - SIH25002**
- **PROBLEM STATEMENT TITLE - Smart Tourist Safety Monitoring & Incident Response System using Al, Geo-Fencing, and Blockchain-based Digital ID**
- **THEME- Travel & Tourism**
- **PS CATEGORY- Software**
- **TEAM ID -**
- **TEAM NAME (Registered on Portal) - QUANTUM LEDGERS**

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IDEA TITLE

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ATITHI BANDHU

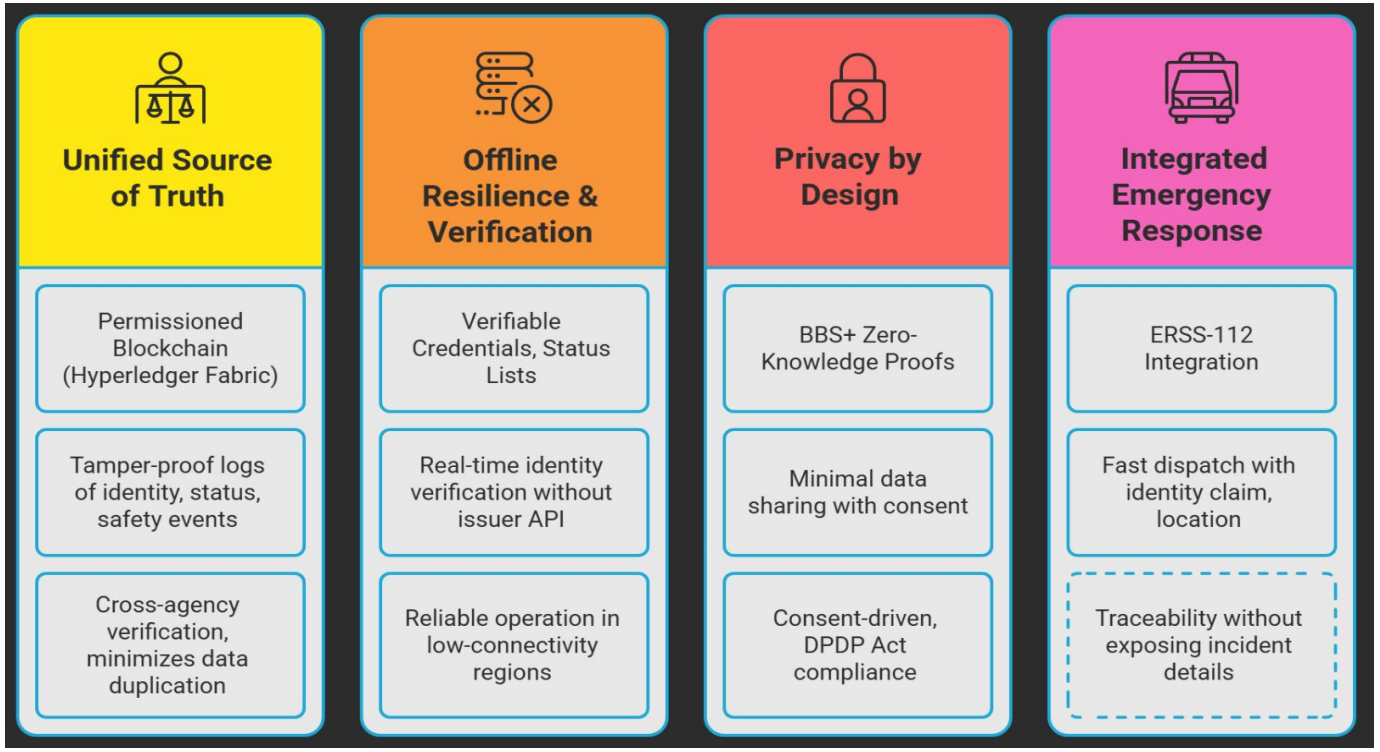
AtithiBandhu: The Digital ID for Every Tourist's Identity & Protection

IDEA/SOLUTION:

Implementation of **Smart Tourist Safety Monitoring & Incident Response System** using AI, Geo-Fencing, and Blockchain-based Digital ID

- Session-based **W3C Tourist Digital ID** issued at arrival of the tourist and closed on departure for secure, time-bound identity management.
- **Secure digital identity** powered by **Hyperledger Fabric**, enabling trusted national-level identity workflows and integration for tourists.
- **AI-driven geofencing and anomaly detection** ensure efficient operation in low-connectivity areas by alerting both authorities and users based on the area's risk score.
- Consent-first, **DPDP-compliant architecture** minimizes data exposure and enforces purpose limitation.

BLOCKCHAIN BASED PROBLEM SOLUTION



Atithi Bandhu is conclusively a secure, privacy-first Smart Tourist Safety System using blockchain, verifiable digital IDs, and AI-powered geofencing to enable real-time safety monitoring, seamless identity verification, and efficient emergency response.

To understand more about our solution via video:Video.mp4

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TECHNICAL APPROACH

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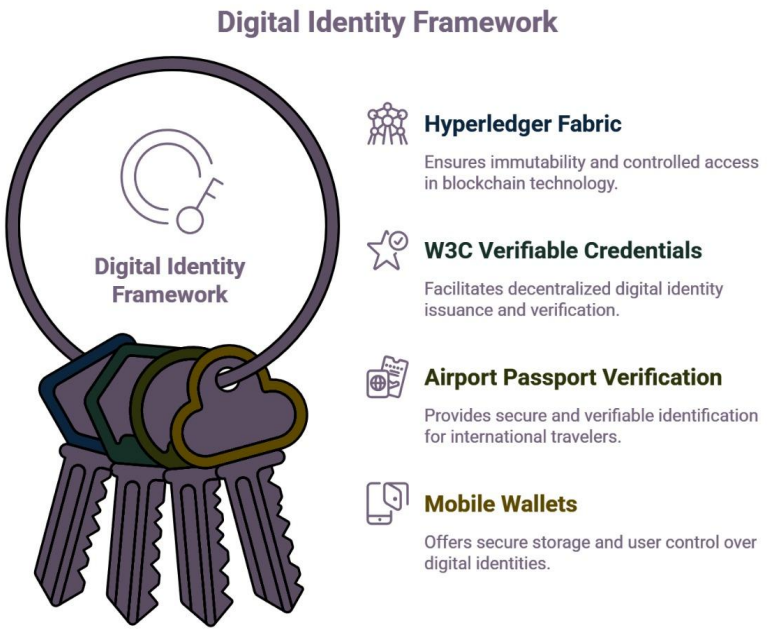
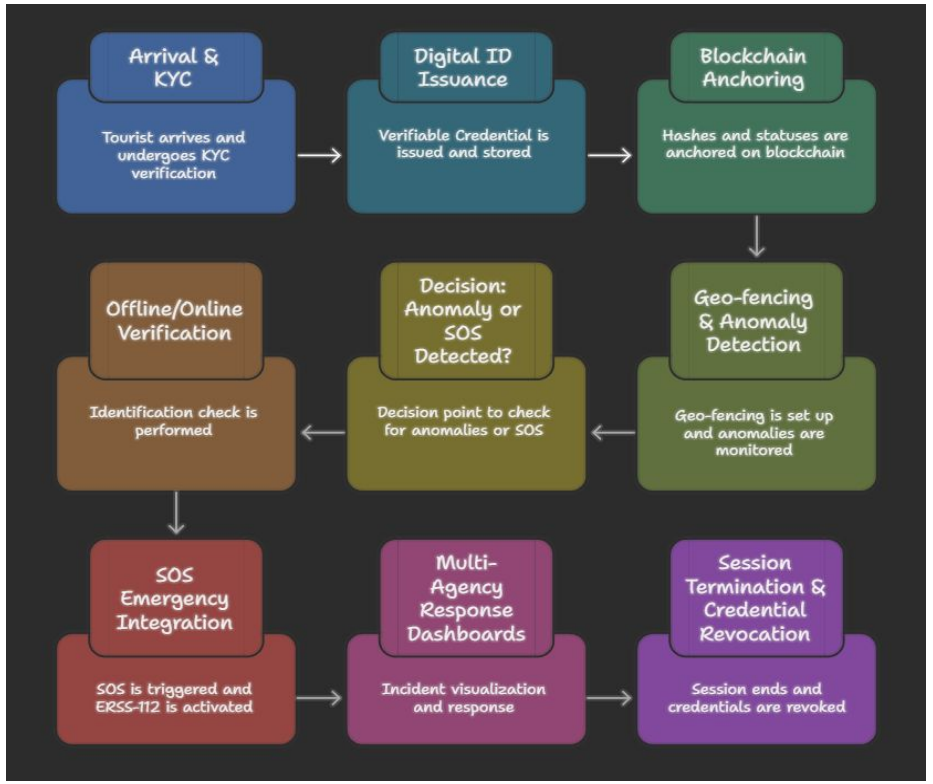
SECURITY FIRST ARCHITECTURE



- The system uses **W3C Verifiable Credentials** to give tourists full control over their digital identity, removing the need for a central authority.
- **Digital identities** are securely stored in **mobile wallets,** allowing users to selectively share verified information when required.
- This architecture is designed for real-world use cases like hotel check-in identity verification or acting as a proof of age, providing a secure, efficient, and privacy-respecting solution.

IMPLEMENTATION FLOWCHART



TECHNOLOGY STACK

W3C
truvera SDK
Hyperledger FABRIC
NEXT.JS
GEOFENCING
node JS
React JS
Flutter

IMPLEMENTATION PROTOTYPE: Tourists receive a digital ID at entry, verified instantly and anchored on blockchain; the app monitors safety and allows offline verification, while SOS alerts and events are logged securely for real-time, private, multi-agency response.

For a complete architecture you may see this diagram:Architecture.png

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FEASIBILITY AND RISK MITIGATION

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Feasibility Challenges:-

- **Constant GPS tracking** drains phone batteries and is unreliable in remote tourist areas.
- Requires **multi-agency collaboration** from Tourism Departments, Police, and District Administration for trusted consortium formation.
- Integrating with diverse, large-scale national systems requires **robust and secure API** connections.

Mitigation:-

- Use **low-power cell tower location** for general tracking and activate high-precision GPS near risk zones, triggering alerts via mobile app or IoT wristband monitoring vitals.
- Since **Hyperledger Fabric** combined with **W3C**'s **BBS+** (for limited data sharing) is powerful, secure, and maintains participant relationships, it's highly scalable, enabling national-level collaboration between organizations without corruption, thanks to its **tamper-proof evidence system**.
- Leverage trusted identity platforms: Integrate via secure APIs to support both **Aadhaar e-KYC** for domestic and **passport-based verification** for international tourists, while connecting to **ERSS-112** for emergencies.

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IMPACT AND BENEFITS

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Impacts :-

- Authorities analyze safety metrics with data driven decisions.
- The government benefits significantly from the use of verifiable digital IDs and privacy-preserving offline identity checks for tourists.

Benefits:-

- AI driven risk assessment and anomaly detection enables safety measures
- Automated digital verification with verifiable credentials reduces time.

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Quantifiable Analysis:-

- Advanced identification and alert systems improves crime mitigation workflows in this domain.
- Increased safety metrics increase tourist trust and allows more economic growth in India

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RESEARCH AND REFERENCES

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Technology	Description	System Usage	Reference
Blockchain Security (Hyperledger Fabric)	Consortium blockchain platform providing tamper-evident storage for status lists, event proofs, and audit logs	Anchors status, SOS, verification, and audit events in a way that agencies can independently validate history and prevent tampering	Mohd NNMSN et al., "Surveys on the Security of Ethereum and Hyperledger Fabric Blockchain Platforms," IJPCC, 2025. https://journals.iium.edu.my/kict/index.php/IJPCC/article/download/526/315/3175
W3C Verifiable Credentials	International standard for digital identity credentials—portable, cryptographically signed, and support selective disclosure	Tourist IDs are issued as verifiable credentials, stored in digital wallets, and used for secure field verification	Zhao Y, Singh J, Kumar N. "A Survey on Decentralized Identifiers and Verifiable Credentials," IEEE Communications Surveys & Tutorials, Jan 2025. https://ieeexplore.ieee.org/document/10503046/
BBS+ Selective Disclosure	Advanced cryptographic methods enabling selective attribute sharing and zero-knowledge proofs	Allows tourists to prove specific claims without exposing their full credential or unnecessary personal information	Bećirović Ramić Š et al., "BLS-MT-ZKP: A novel approach to selective disclosure of claims from digital credentials," arXiv, Aug 2024. https://arxiv.org/html/2402.15447v3
AI/ML Anomaly Detection	On-device and backend algorithms for anomaly detection and geofence monitoring	Detects risky movements, route deviations, prolonged inactivity, and crowding; triggers real-time safety alerts	Vishwakarma A, Sharma N. "Implementing Real-Time Anomaly Detection in Mobile Security Logs," IJRASET, April 2025. https://doi.org/10.22214/ijraset.2025.68272
Digital Wallets & Mobile Security	Mobile application and secure device storage for digital credentials and personal data	Ensures personal information remains off-chain and under user control, providing privacy and minimizing exposure	Miller J, Thompson K. "A Complete Guide to Digital Wallet Security," JPLoft Technology, March 2025. https://www.jploft.com/blog/digital-wallet-security

RESEARCH AND REFERENCES

Technology	Description	System Usage	Reference
Dual-Polarization SAR Vegetation Index (DPSVI)	Radar-based vegetation index using Sentinel-1 VV/VH bands to penetrate clouds, smoke, and rain.	Used to create all-weather vegetation masks, allowing the system to monitor forest cover 24/7 even during monsoon seasons.	*DOI: 10.1080/22797254.2020.1867901* (Validated accuracy of 93% in vegetation masking)
Prithvi EO 2.0 (Vision Transformers)	NASA-IBM's 300M parameter foundation model designed for multi-spectral Earth observation data.	Fine-tuned to classify specific tree species (Oak, Pine, etc.) and determine forest health by fusing optical and SAR data.	*NASA-IBM & TreeSatAI Dataset:* Validated on 50,381 patches across 20 European tree species.
MegaDetector v6 (Edge AI)	YOLO v10-based object detection model optimized for identifying animals in camera trap imagery	Deployed on Raspberry Pi 4 + Coral TPU for real-time species detection, filtering out 95% of empty images before transmission.	*LILA BC Benchmark:* Achieves 95.7% animal detection accuracy; adapted from Wildlife Insights methodologies
XGBoost Predictive Modeling	Scalable machine learning classifier used for analyzing large-scale structured data (weather, topography, fuel load).	Powers the "Risk Prediction Engine" to forecast fire ignition probability 72 hours in advance based on fuel moisture and micro-climate	Validated on 10 years of MODIS fire pixel data and California historical fire records (87% precision).
Zarr & Cloud Optimized GeoTIFFs (COGs)	Cloud-native geospatial data formats optimized for high-speed parallel querying of massive multidimensional arrays.	Enables the "Time Machine" feature, allowing sub-second querying of 7+ years of satellite history for trend analysis.	*IEEE JSTARS Submission:* "Multi-Sensor Fusion for All-Weather Tree Species Classification" (Under Review).